

PRODUCT REQUIREMENTS DOCUMENT

Document Version	1.0
Product Name	Palm Oil Ripeness Detection
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Stakeholder	Eric Julianto
Doc Stage	Done
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Overview

The Palm Oil Detection project aims to enhance agricultural practices by leveraging computer vision technology. The focus is on detecting the maturity level of palm oil using YOLOv8, a state-of-the-art object detection algorithm. six maturity classes are considered: unripe, underripe, ripe, overripe, abnormal, empty bunch.

Implementation includes the development of a Command Line Interface (CLI) for image-based maturity detection. Users can input an image, and the system will output predictions for the maturity class of the palm oil in the image.

Additionally, the project supports real-time processing of video streams.Object counting is implemented to determine the number of palm trees passing through the stream, along with their respective maturity levels. The system allows for both internet and local connectivity with Real-Time Streaming Protocol (RTSP) for seamless data transmission.

Future plans involve the integration of ESP32 Cam, a low-cost, small-sized camera module, into the system. This hardware enhancement is aimed at capturing images for maturity detection directly from the field. The collected data will be sent to a cloud server centralized processing and analysis.

Problem Statement

The existing methods for assessing palm oil maturity lack efficiency and automation. Manual processes lead to inaccuracies, delays, and a lack of centralized control. Key issue:

Inability to obtain a comprehensive palm oil maturity across different plantations.

increased manual effort in coordinating maturity assessments and monitoring across various locations.

Objective

- The primary objective is to create an integrated solution for palm oil maturity detection and monitoring.
- Specific goals include:
 - Develop a CLI for users to easily detect maturity detection on individual images.
 - Implement object counting in video streams to track the number of palm and their maturity levels.
 - Integrate ESP32 Cam into the system for on-field image capture.
 - Establish connectivity with a cloud server for centralized data processing and analysis.

Key Objectives:

- Streamlined Maturity Detection:** Centralize and automate the maturity detection process using YOLOv8, ensuring accuracy and efficiency.
- Real-time Monitoring:** Provide real-time insights into the maturity levels of palm oil video streams, enabling prompt decision-making.
- Hardware Integration:** Incorporate ESP32 Cam for field-level image capture, expanding the scope of data collection.
- Cloud Connectivity:** Establish a connection with a cloud server for centralized data processing, enabling scalability and remote access.

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	Tags	Guidelines
Decider	Eric Julianto	Ensures the successful implementation and optimization of the Palm Oil Detection project, making critical decisions related to the integration of YOLOv8, ESP32 Cam, and cloud server functionalities. Examples include determining the model parameters for maturity classification and approving the final deployment strategy.
Accountable		Managers the overall project execution, overseeing the deployment, testing and deployment phases. Responsible for coordinating team efforts, ensuring that milestones are met, and resolving any obstacles hindering the project's progress.

	Tags	Guidelines
Responsible		Actively engages in the hands-on tasks related to the development and implementation of the CLI, image processing, and video stream analysis. Takes ownership of the technical aspects, working closely with the development team to address challenges and ensure successful feature delivery.
Consulted		Collaborates with stakeholders to gather insights and refine the user interface of the CLI. Actively seeks feedback from end-users and domain experts to enhance the overall effectiveness of the palm oil detection system.
Informed		Maintains open communication challenges, and opportunities related to the project. Actively progress, challenges, and opportunities related to projects. Actively seeks and incorporates feedback from both technical and non-technical perspectives to refine and improve the solution.

Project Timeline

Time Period	Activity	PIC
8 Jan	Grooming	PM
15 Jan-26 Jan	System Design and Architecture	Team Engineering
26 Jan-9 Feb	CLI Development and Image Processing	Team Engineering
9 Feb-12 Feb	Implement Flask	Team Engineering
12 Feb-16 Feb	Finishing	Team Engineering

Success Metrics

Metrics	Definition	Actual	Target
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Metrics	Definition	Actual	Target
Maturity Detection Accuracy	% of correctly classified palm oil maturity levels.	75% accuracy.	Achieve a maturity detection of 90%.
CLI Usability	User satisfaction with the ease of using the Command Line Interface (CLI) for image-based detection.	80% user satisfaction.	80% user satisfaction.
ESP32 Cam Integration Success	Successful integration and functionality of ESP32 Cam for on-field image capture.	Integration Completed.	Achieve a fully functional integration with ESP32 Cam.
Cloud Server Connectivity Reality	Reliability of the connection between the system and the cloud server.	95% reliability.	Achieve 99% connectivity reliability.
User Feedback Incorporation	% of user feedback actively incorporated into system improvements.	70% feedback integration.	Achieve 90% feedback incorporate rate.
System Performance Enhancement	Improvement in overall system performance during image and video processing.	60% improvement.	Achieve an 80% improvement in system performance.

User Stories

Title	User Story	Acceptance Criteria	Priority
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Title	User Story	Acceptance Criteria	Priority
Maturity Detection Accuracy Enhancement	As a user, I want to instantly and accurately determine the ripeness of my palm fruits using my phone or a dedicated camera, so i can optimize harvest timing and maximize yield.	Given i have a bunch of palm oil fruits, When i capture a photo using my smartphone app or the ESP32 Cam, Then the system analyzes the fruit characteristics (color, texture, etc.) and provides a clear prediction : Unripe, Underripe, Ripe, Overripe, Abnormal, Empty bunch.	High
Capture On-Field Seamlessly with ESP32 Cam	As a user, I want to easily collect maturity data from various locations in my plantation without manual effort, so I can monitor overall fruit development and make informed harvesting decisions.	Given the ESP32 Cam is setup in strategic locations, When I configure the ESP32 Cam to capture bunch images at predetermined intervals, Then the system automatically captures images and securely	High

Title	User Story	Acceptance Criteria	Priority
Centralized Data Management with Reliable Cloud Connectivity	As a user, I want to access and analyze palm oil maturity data from all plantations in one central location, so I can optimize harvesting across regions and make strategic business decisions.	Given the system maintains a reliable connection (at least 98%) to the cloud server, even in areas with weak internet, When I access the user-friendly web interface. Then I can see data from all plantations securely stored and easily accessible.	High

References

[Link module](#)